Ali Mortazavi

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Summary

PhD candidate in online learning and game theory. Seeking Applied Scientist and research roles in ML and optimization.

Education

University of Victoria — Victoria, BC, Canada
Ph.D. in Computer Science, GPA: **9/9** — Jan 2021 – Dec 2026 (Expected)

- Research focus: Game Theory, Sequential Prediction, Adaptive Machine Learning

Amirkabir University of Technology — Tehran, Iran
M.Sc. Artificial Intelligence — Sept 2017 – Sept 2020
B.Sc. Computer Software Engineering — Sept 2013 – Sept 2017

Research Experience

PhD **Research Assistant** — Jan 2021 – Present
Machine Learning Theory Group, University of Victoria

- **Reputation-Based Sequential Forecasting** `(Game Theory, Information Elicitation)`
  - Co-designed a sequential algorithm that combines predictions from many forecasters over time. It is designed with a strategy-proof guarantee: it is in each forecaster's best interest to report their beliefs truthfully, even when they care about their long-term reputation.
  - Contributed to a mathematically rigorous proof showing that the algorithm's error is as good as the best forecaster's as $T \to \infty$ (more precisely, the algorithm's error converges to the best forecaster's error at roughly the rate $1/\sqrt{T}$). This result resolved an open question in the field.
  - Extended the results to the partial information setting, where the algorithm only observes the outcome of the forecaster it follows rather than all forecasters (a multi-armed bandit problem), the first such result for this setting.
- **Aggregating Advice from Selection-Seeking Forecasters** `(Incentives, Partial feedback)`
  - Investigated and identified the limitation of the only known (at the time) strategy-proof sequential forecaster selector algorithm, resolving whether its slower rate was an artifact of the analysis.
  - Used martingale concentration arguments to prove the inherent performance limitation of the algorithm.
- **Adaptive Decision-Making While Preserving Group Fairness** `(FTRL, Fairness)`
  - Contributed to identifying a class of algorithms with mathematically provable guarantees on preserving fairness without prior knowledge of demographic group sizes.
  - Characterized the environment in which the decision maker achieves its best possible performance relative to the best expert, showing our bound is tight.

Publications

- Junpei Komiyama, Nishant Mehta, and Ali Mortazavi. "No-Regret Incentive-Compatible Online Learning under Exact Truthfulness with Non-Myopic Experts" *ACM Conference on Economics and Computation*, 2025.
- Ali Mortazavi, Junhao Lin, and Nishant Mehta. "On the Price of Exact Truthfulness in Incentive-Compatible Online Learning with Bandit Feedback" *International Conference on Artificial Intelligence and Statistics*. PMLR, 2024.
- Cristóbal Guzmán, Nishant Mehta, and Ali Mortazavi. "Best-Case Lower Bounds in Online Learning" *Advances in Neural Information Processing Systems*, 34 (2021).

Technical Skills

- **Mathematical Methods:** Linear and convex optimization (LP formulation, duality, KKT), Online learning, Multi-armed bandits and adaptive experimentation, Probability and concentration inequalities, Game theory, mechanism and auction design
- **Programming Languages:** Python, Java, C++
- **Frameworks & Libraries:** NumPy, Pandas, Matplotlib, TensorFlow, scikit-learn, PyTorch
- **Tools:** Git, LaTeX

Applied Machine Learning Experience

- **Image Denoising and Segmentation** (Report, GitHub) `(Python, Image Processing)`
  Optimized image denoising and segmentation with Simulated Annealing and Markov Random Fields. Compared performance across HSV, RGB, and Grayscale color spaces.
- **Text Summarization** (Report, GitHub): `(Python, TensorFlow, NLP)`
  Developed a graph-based text summarization method using word2vec and PageRank. Evaluated the performance with word-level and sentence-level representations using ROUGE metrics.

Internship

Shanghai University of Finance and Economics Shanghai, China
Research Internship at Institute for Theoretical Computer Science August 2019 – September 2019

- Project: Applied various approximation and probabilistic techniques to measure the performance of a novel Online Stochastic Matching algorithm.

Teaching Experience

Teaching Assistant Jan 2021 - Present
Computer Science Department University of Victoria

- **Notable Responsibilities**: Designed and taught labs and tutorials, helped prepare new TAs with teaching tasks
- **Courses**: Algorithms and Data Structure II, Theory of Computation, Data Mining, Collective Decision-Making, Advanced Data Structure and Optimization, Machine Learning Theory.

Awards

- AISTATS 2025 Best Reviewer Award (Top 4% reviewers) 2025
- University of Victoria Graduate Awards for Top-Performing Students 2021-2024
- University of Victoria Graduate TA Award 2021-2024
- Donald Wagg Graduate Scholarship 2024
- Alexander and Helen Stafford MacCarthy Muir Graduate Scholarship 2024
- Jarmila Vlasta Von Drak Thouvenelle Graduate Scholarship 2024
- Charles S. Humphrey Graduate Student Award 2022
- UVic PhD Fellowship Award 2021-2022
- Ranked 3rd (out of 100) in terms of Cumulative GPA among students of computer engineering, 2013 Entrance 2017
- Awarded direct admission to the M.Sc. program in Artificial Intelligence at Amirkabir University of Technology as a Talented Undergraduate Student 2017

Presentations

- How to sequentially pick the best forecaster when everyone is competing for reputation, *Excellence in Graduate Research Symposium*, University of Victoria, 2025
- Best-case lower bounds in online learning, *Advances in Neural Information Processing Systems (NeurIPS)*, 2021

Volunteer Activities

Reviewer for ICML 2025, AISTATS 2025, NeurIPS 2024